Drift Assignment — Project Setup & Baseline Delta

> **Purpose**: Show the interviewer *exactly* what was added on top of an empty Unity 6 URP 3D template — nothing hidden, nothing hand-waved. This doc pairs with REQUIREMENTS.md and gets exported to PDF as part of the final deliverable bundle.

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1. Baseline — What a fresh URP 3D Sample template (Unity 6.0.69f1) ships with

A brand-new URP 3D project on Unity 6 contains:

Packages (top-level)

- `com.unity.render-pipelines.universal` — URP 17.0.4
- `com.unity.inputsystem` — 1.19.0 (new Input System)
- `com.unity.ide.rider`, `com.unity.ide.visualstudio` — IDE integrations
- `com.unity.collab-proxy` — Version Control integration
- `com.unity.test-framework`, `com.unity.timeline`, `com.unity.ugui`
- `com.unity.ai.navigation`, `com.unity.multiplayer.center`, `com.unity.visualscripting`
- Built-in modules (physics, animation, audio, vehicles, terrain, etc.)

Assets

- `Assets/Scenes/SampleScene.unity`
- `Assets/Settings/` — `PC_RPAsset`, `PC_Renderer`, `Mobile_RPAsset`, `Mobile_Renderer`, `DefaultVolumeProfile`, `SampleSceneProfile`, `UniversalRenderPipelineGlobalSettings`
- `Assets/TutorialInfo/` — sample icons, layout, tutorial scripts (template clutter)
- `Assets/Readme.asset` — welcome sheet (template clutter)
- `Assets/InputSystem_Actions.inputactions` — default input actions

Player settings defaults

- Build target: **Windows Standalone** (PC)
- Scripting backend: **Mono**
- API compatibility: .NET Standard 2.1
- Company name: **DefaultCompany**
- Product name:

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2. Current state — What's already in *this* project

Verified against `Packages/manifest.json`, `Packages/packages-lock.json`, and `ProjectSettings/ProjectSettings.asset` at the time of writing.

Packages added on top of URP template ✅

Package	Version	Purpose
`com.coplaydev.unity-mcp`	git (`main` branch)	**Agentic Editor automation via Claude / MCP**
`com.unity.cinemachine`	3.1.7	Multi-angle car cameras (chase / hood / cinematic / look-back)
`com.unity.memoryprofiler`	1.1.12	Deep memory profiling on-device (Phase 9 optimization)
`com.unity.mobile.android-logcat`	1.4.7	View device logs inside Unity Editor for Android debugging
`com.unity.recorder`	5.1.6	Screen / gameplay recording for demo videos + replay tie-ins
`com.unity.splines`	2.8.3	(Transitive dep of Cinemachine; useful for road / camera paths later)
`com.unity.2d.sprite`	1.0.0	Sprite editor for HUD icon atlasing (Phase 4 / 9)

Player settings already changed ✅

Setting	Baseline	This project	Notes
Product name	`<template>`	**DriftAssignment**	Set
Company name	DefaultCompany	**GrindingStudio**	✅ set
Android scripting backend	Mono (Standalone default)	**IL2CPP**	✅ Play Store requirement met
Android target architectures	ARMv7	**ARM64 only** (bitmask `2`)	✅ Play Store 64-bit requirement met
API compatibility level	.NET Standard 2.1	**.NET Standard 2.1** (`6`)	✅
Android min SDK	23 (Android 6.0)	**24 (Android 7.0)**	✅ per REQUIREMENTS spec
Android target SDK	Highest installed	**Auto (** `0` **)**	✅
Active build target	Windows Standalone	*(inferred Android — `scriptingBackend.Android` configured)*	Verify in Editor

Assets currently present

```
Assets/
  Scenes/Main.unity                    ✅ renamed from SampleScene
  Settings/
    Mobile_RPAsset.asset               ✅ mobile-tuned (HDR off, MSAA 2x, ShadowDist 40)
    Mobile_Renderer.asset
    PC_RPAsset.asset                   (unused on Android; kept for Editor)
    PC_Renderer.asset
    DefaultVolumeProfile.asset
    SampleSceneProfile.asset
    UniversalRenderPipelineGlobalSettings.asset
  InputSystem_Actions.inputactions     (baseline — extend with driver actions in Phase 2/4)
  _Project/                            ✅ full folder tree (Scripts, Prefabs, SOs, Materials, etc.)
    Scripts/{Core,Vehicle,Damage,Input,UI,Camera,Character}/
      each with DriftAssignment.<Module>.asmdef ✅
  ThirdParty/
```

```
    Kenney/LICENSE.txt                ✅
    Quaternius/LICENSE.txt            ✅
    Mixamo/LICENSE.txt                ✅
  (removed) TutorialInfo/             ✅
  (removed) Readme.asset              ✅
```

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3. Delta table — what's added / removed / changed on top of the URP template

Legend: 🟢 done | 🟡 partially done | 🔴 pending

3.1 Packages

Change	Item	Status	Phase
Add	Cinemachine 3.1.7	🟢	0
Add	CoplayDev Unity MCP (git)	🟢	0
Add	Memory Profiler	🟢	0
Add	Android Logcat	🟢	0
Add	Unity Recorder	🟢	0
Add	2D Sprite	🟢	0
Add	(Transitive) Splines	🟢	0

3.2 Build & Player settings

Change	Item	Target value	Status	Phase
Change	Active build target	Android	🟡 (scripting backend set — confirm `Switch Platform` was clicked in Editor)	0
Change	Android scripting backend	IL2CPP	🟢	0
Change	Android target architectures	ARM64 only	🟢	0
Change	API compatibility	.NET Standard 2.1	🟢	0
Change	Android min SDK	24 (Android 7.0)	🟢	0
Change	Company name	GrindingStudio	🟢	0
Change	Orientation	Landscape (auto rotate: Landscape L + R)	🟢 landscape L+R on, portrait off	0
Change	Multi-touch enabled	true	🟡 default enabled with Input System — verify in Editor	0

3.3 URP asset tuning (Mobile_RPAsset)

Change	Item	Target	Status	Phase
Change	HDR	Off	🟢 `m_SupportsHDR: 0`	0
Change	MSAA	2×	🟢 `m_MSAA: 2`	0
Change	Shadow distance	40 m	🟢 `m_ShadowDistance: 40`	0
Change	Shadow cascades	1 (single)	🟢 already `1`	0
Change	Depth texture	Off (turn on only if needed for water/decals)	🟢 already off	0
Change	Opaque texture	Off (mobile perf)	🟢 already off	0
Change	Post-processing	On (mild bloom only)	🔴 configure in Phase 8	0 / 8

3.4 Project folders & scenes

Change	Item	Status	Phase
Remove	`Assets/TutorialInfo/`	🟢	0
Remove	`Assets/Readme.asset`	🟢	0
Rename	`Scenes/SampleScene.unity` → `Scenes/Main.unity`	🟢 (+ EditorBuildSettings updated, enabled=1)	0
Add	`Assets/_Project/Scripts/{Core,Vehicle,Damage,Input,UI,Camera,Character}/`	🟢	0
Add	`Assets/_Project/Prefabs/{Vehicles,Environment,UI}/`	🟢	0
Add	`Assets/_Project/ScriptableObjects/{TuningPresets,CarConfig}/`	🟢	0
Add	`Assets/_Project/Materials` , `Textures` , `Audio` , `Fonts`	🟢	0
Add	`Assets/ThirdParty/{Kenney,Quaternius,Mixamo}/` with `LICENSE.txt`	🟢	0 / 1
Add (Phase 1)	`Assets/ThirdParty/{PolyHaven,Sketchfab,PolyPizza}/` with `LICENSE.txt` — as needed	🔴 pending	1
Add	Assembly Definitions per module (`DriftAssignment.*`)	🟢 all 7 created	0

3.5 Repo-level docs & tooling

Change	Item	Status	Phase
Add	`.editorconfig` (mirrors Unity C# style guide)		0
Add	`Doc/REQUIREMENTS.md`		Docs
Add	`Doc/ARCHITECTURE.md`		Docs
Add	`Doc/IMPLEMENTATION_LOG.md`	scaffold	Docs
Add	`Doc/OPTIMIZATION.md`	scaffold	Docs
Add	`Doc/CREDITS.md`	scaffold	Docs
Add	`Doc/PROJECT_SETUP.md` (this doc)		Docs
Add	`Doc/build-pdfs.ps1` (Pandoc / md-to-pdf pipeline)		Docs
Add	`CLAUDE.md` at project root	pending (after Phase 0 lands, next up)	11

3.6 Unity MCP registration (already installed, verify wired to Claude)

Change	Item	Status
Verify	MCP server responds on `http://127.0.0.1:8080/mcp`	verify with Unity Editor open
Verify	`claude mcp list` shows `UnityMCP`	verify
Register (if missing)	`claude mcp add --scope local --transport http UnityMCP http://127.0.0.1:8080/mcp`	(see REQUIREMENTS §3)

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4. Phase 0 execution checklist — status

#	Task	Status
1	Active build target = Android	🟡 verify `Switch Platform` clicked in Editor
2	`AndroidMinSdkVersion` 23 → 24	🟢
3	Company name set	🟢 GrindingStudio
4	Orientation = Landscape L+R only	🟢
5	Multi-touch enabled	🟡 default with Input System — verify in Editor
6	`Mobile_RPAsset` mobile tune (HDR, MSAA, ShadowDist, cascades, depth/opaque)	🟢
7	Delete `TutorialInfo/` and `Readme.asset`	🟢
8	Rename `SampleScene.unity` → `Main.unity` (+ update EditorBuildSettings)	🟢
9	Create `_Project/` folder tree	🟢
10	Create `ThirdParty/` folder tree + `LICENSE.txt`	🟢
11	Create 7 Assembly Definitions	🟢
12	Unity MCP: `claude mcp list` + live health-check	🔴 pending — run `claude mcp add ...` per REQUIREMENTS §3
13	Empty Android build produces APK	🔴 pending — user action in Editor

Two items still need user action in the Editor:

- Open Unity → `File → Build Profiles → Android → Switch Platform` (if not already active). This triggers Unity to reimport assets for Android and confirms Phase 0.
- Run `claude mcp add --scope local --transport http UnityMCP http://127.0.0.1:8080/mcp` (or click **Configure** in the CoplayDev Unity MCP window) so Claude can drive the Editor for Phases 1+.
- Attempt a headless Android build (`File → Build`) to confirm APK produces cleanly.

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5. What we did *not* need to add

Called out explicitly because interviewers often ask "did you remember X?":

- **URP** — already in the template
- **Input System** — already in the template (v1.19.0)
- **Vehicles module** (WheelCollider lives here) — built-in, already available
- **Physics module** — built-in, always available
- **Terrain module** — built-in (used for dune flanks)
- **UI Toolkit / UGUI** — UGUI is in template; using UGUI for HUD (Screen Space Overlay)

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6. Rationale for each package we added

Package	Why we added it
Cinemachine	Multiple camera angles (chase / hood / cinematic / look-back) matching the reference camera-cycle button. Blends between cams are free with Cinemachine and a lot of manual math to replicate.
CoplayDev Unity MCP	Lets Claude Code drive the Unity Editor directly — create GameObjects, wire prefabs, run tests, without leaving the chat. Massive iteration-speed multiplier for this assignment.
Memory Profiler	Phase 9 optimization requires on-device memory snapshots. Deep Profiler alone isn't enough; the Memory Profiler package captures heap detail.
Android Logcat	View `Debug.Log` output from the device inside the Unity Editor without needing external `adb logcat` sessions.
Unity Recorder	Capture gameplay video for demo submission + potential replay-record feature (stretch goal per REQUIREMENTS §4.1).
2D Sprite	Needed to build sprite atlases for HUD icons (Phase 4 / 9 optimization — atlasing collapses UI draw calls).

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7. What this table means for the interviewer

- **No hidden magic**: everything the project has beyond the URP template is listed, versioned, and justified.
- **Nothing added for its own sake**: each package traces to a specific phase or requirement.
- **Reversible**: removing any single addition and doing that work by hand is possible — the choices are pragmatic, not architectural.